Forecasting Performance

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2025-11-24

In this example, we compare the forecasting performance of all the non-seasonal methods.

The data livestock {fpp2} spans the period 1961—2007.

We withhold the period 2001—2005 as a test set, and use the data from 1970 up to and including year 2000 for the training set.

```r
library(fpp2)
```

```
## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo
```

```
## -- Attaching packages ---------------------------------------------- fpp2 2.5 --
```

```
## v ggplot2   4.0.1      v fma       2.5
## v forecast  8.24.0     v expsmooth 2.3
```

```
##
```

```r
library(forecast)
data(livestock)
livestock
```

```
## Time Series:
## Start = 1961
## End = 2007
## Frequency = 1
##  [1] 232.2890 229.5363 233.1459 243.7637 252.6029 259.6774 260.7669 269.7841
##  [9] 266.4150 263.9177 268.3072 260.6626 266.6394 277.5158 283.8340 290.3090
## [17] 292.4742 300.8307 309.2867 318.3311 329.3724 338.8840 339.2441 328.6006
## [25] 314.2554 314.4597 321.4138 329.7893 346.3852 352.2979 348.3705 417.5629
## [33] 417.1236 417.7495 412.2339 411.9468 394.6971 401.4993 408.2705 414.2428
## [41] 407.9980 403.4608 413.8249 428.1050 445.3387 452.9942 455.7402
```

1. Subset of data starting at 1970 and ending at 2000

We extract the subset for the training (1970-2000) and test (2001-2005) sets.

```r
train <- window(livestock, start = 1970, end = 2000)
test <- window(livestock, start = 2001, end = 2005)
```

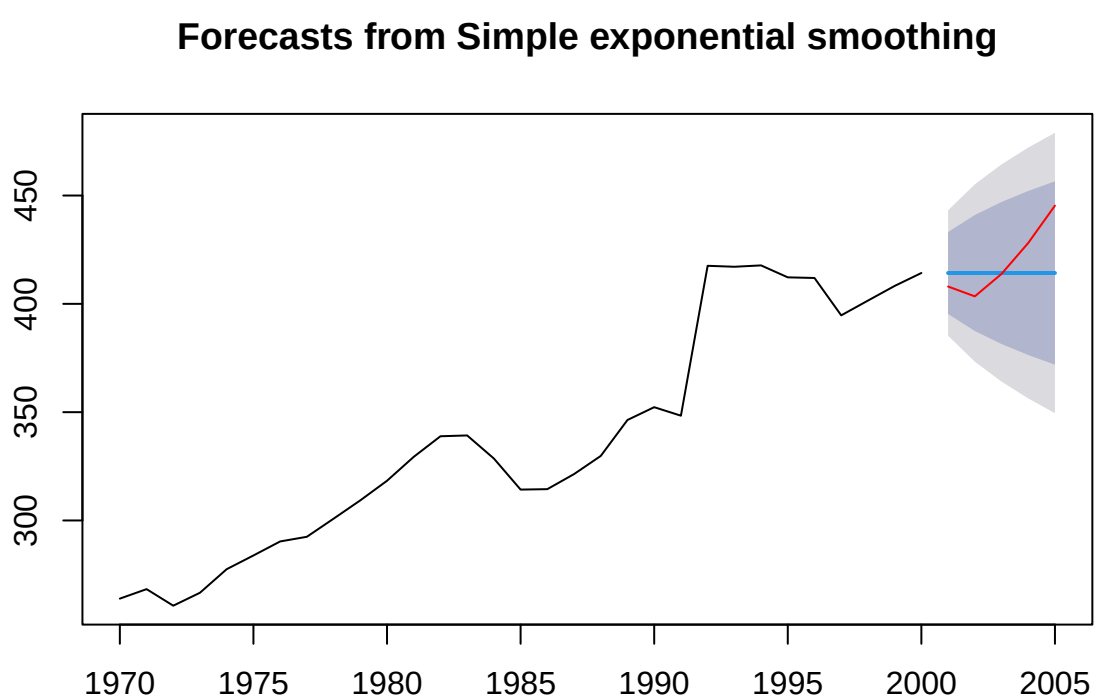
2. Estimation by ses {forecast}, accuracy {forecast} in the training and test sample, and level plot

```r
#Simple exponential smoothing
train_ses <- ses(train,initial="simple", h = 5) #estimated alpha #h=5 (2005-200=5)

#Accuracy on training and test sets
accuracy(train_ses, test)
```

```
##                     ME     RMSE       MAE      MPE     MAPE      MASE
## Training set 4.849195 14.76849  8.723551 1.374734 2.519725 0.9677419
## Test set     5.502688 16.21444 12.480553 1.183323 2.904903 1.3845227
##                      ACF1 Theil's U
## Training set -0.02507143        NA
## Test set      0.38875011  1.405123
```

```r
plot(train_ses)
lines(test,col="red")
```



Interpretation

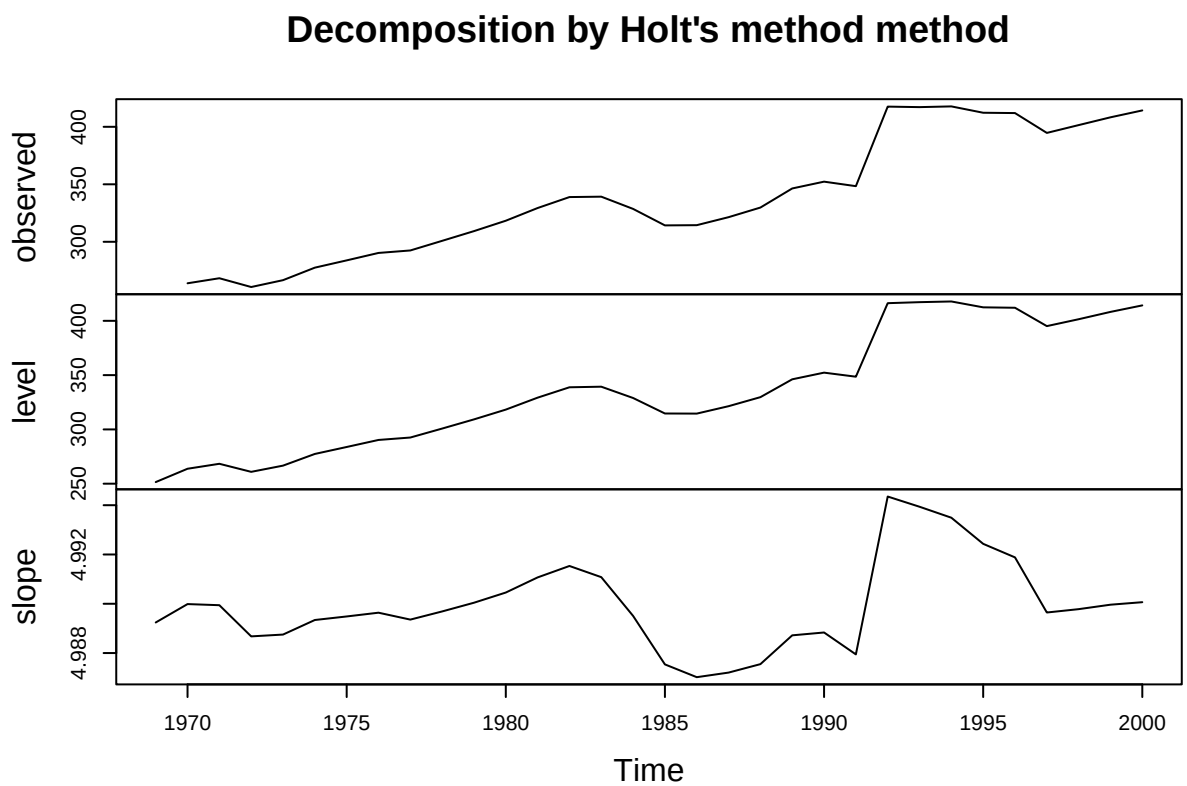
Comparing the SeS forecast, the blue line, with the Test sample, the red line, we clearly see the difference: test sample has a slight downward movement, and then a rapid positive increase, nothing alike the forecast, which is constant.

3. Estimation by holt {forecast}, accuracy {forecast} in the training and test sample, and level and slope plot

```
train_holt <- holt(train)
names(train_holt)
```

```
##  [1] "model"     "mean"      "level"     "x"         "upper"     "lower"
##  [7] "fitted"    "method"    "series"    "residuals"
```

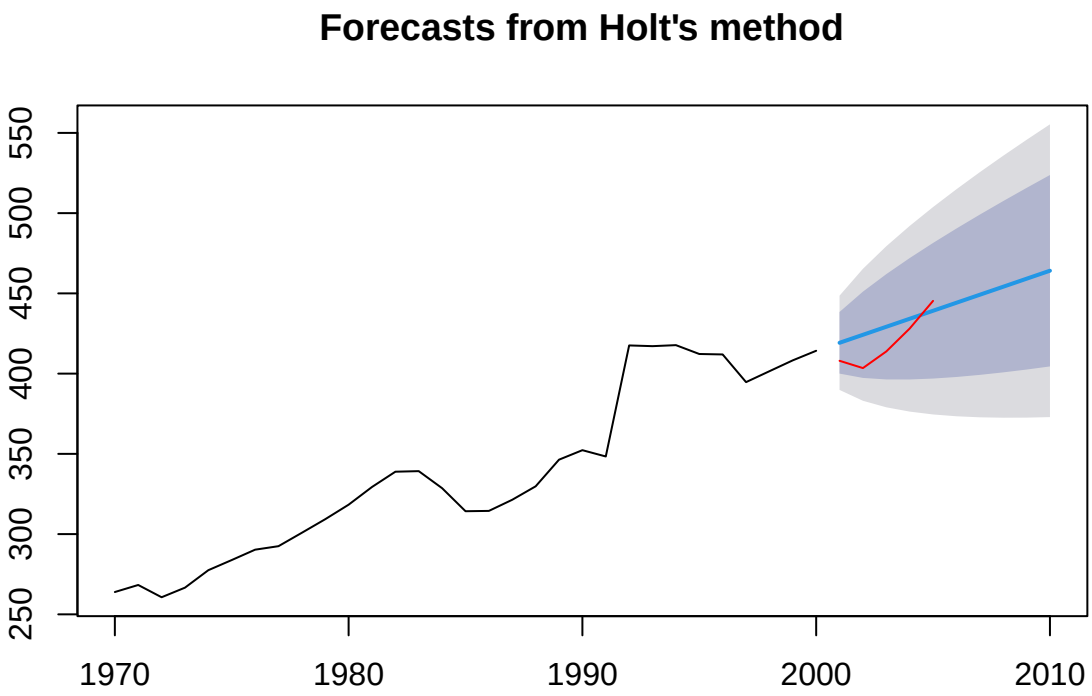
```
plot(train_holt$model)
```




```
accuracy(train_holt, test)
```

```
##                      ME     RMSE       MAE         MPE     MAPE      MASE
## Training set  0.2657316 13.98205  7.808464  0.01732775 2.222262 0.8662273
## Test set     -9.4468205 13.16874 11.913343 -2.32758799 2.881441 1.3215996
##                      ACF1 Theil's U
## Training set -0.006192567        NA
## Test set      0.283914721  1.113641
```

```
plot(train_holt)
lines(test, col = "red")
```



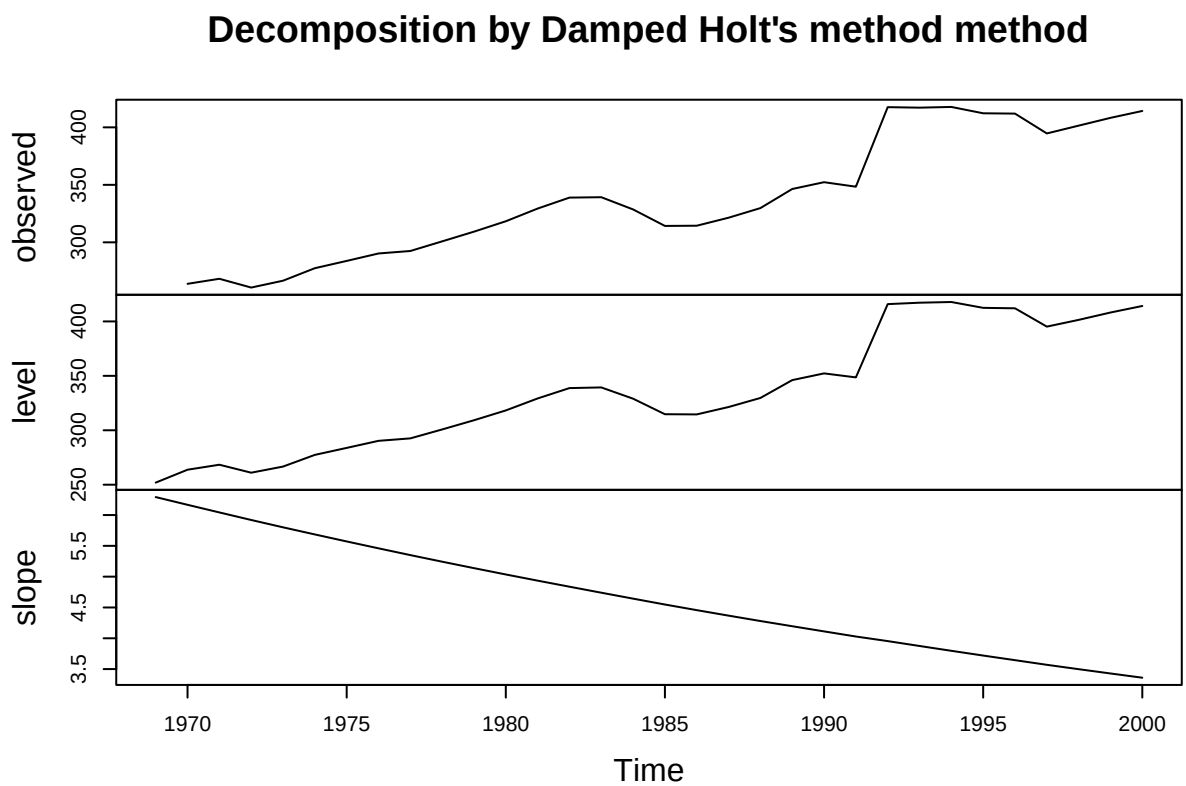
Interpretation

Comparing the Simple Holt forecast, the blue line, with the Test sample, the red line, we observe a slight difference, smaller than with SeS forecast. Now, the forecast, is not constant, but upwarded, with increasing tendency matching the test's, but with different slope. At year 2004-2005, the difference between test sample and Holt forecast seems the smallest.

4. Estimation by holt (damped=TRUE) {forecast}, accuracy {forecast} in the training and test sample, and level and slope plot

$ETS(A, A_d, N)$, equivalent to ARIMA(1,1,2)

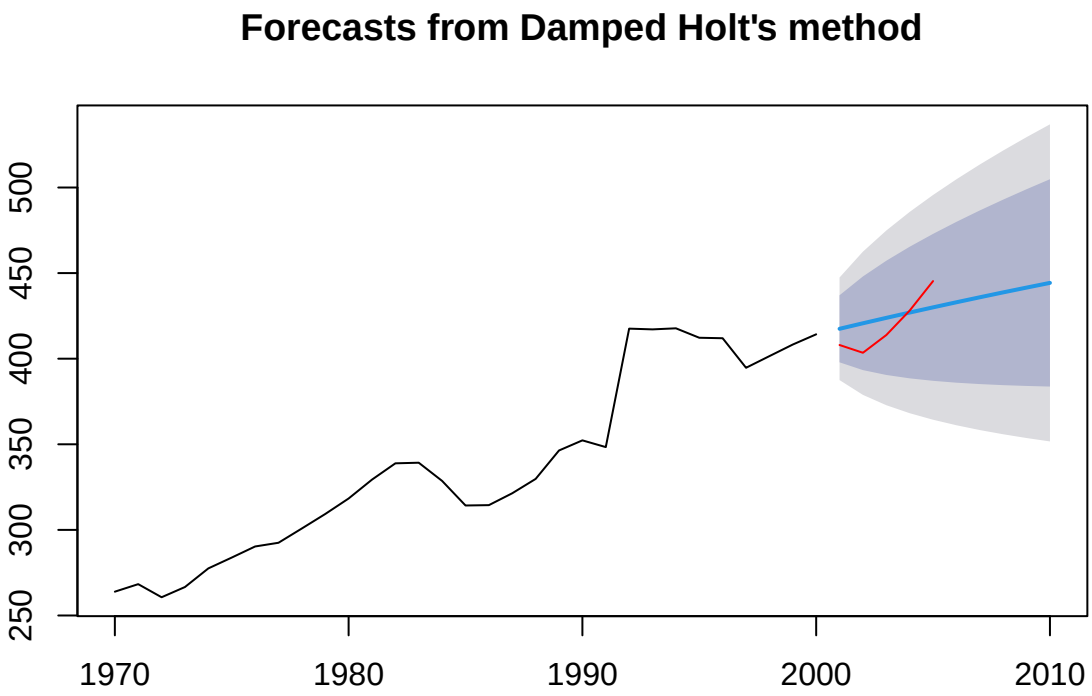
```
train_holt_damped <- holt(train, damped = TRUE)
plot(train_holt_damped$model)
```




```r
accuracy(train_holt_damped,test)
```

```
##                      ME     RMSE       MAE         MPE     MAPE      MASE
## Training set  0.6250276 13.99691  7.743533  0.08683575 2.200726 0.8590242
## Test set     -4.0465767 12.03438 10.646274 -1.06012383 2.546235 1.1810381
##                     ACF1 Theil's U
## Training set 0.001678999        NA
## Test set     0.342709801  1.013854
```

```r
plot(train_holt_damped)
lines(test, col = "red")
```

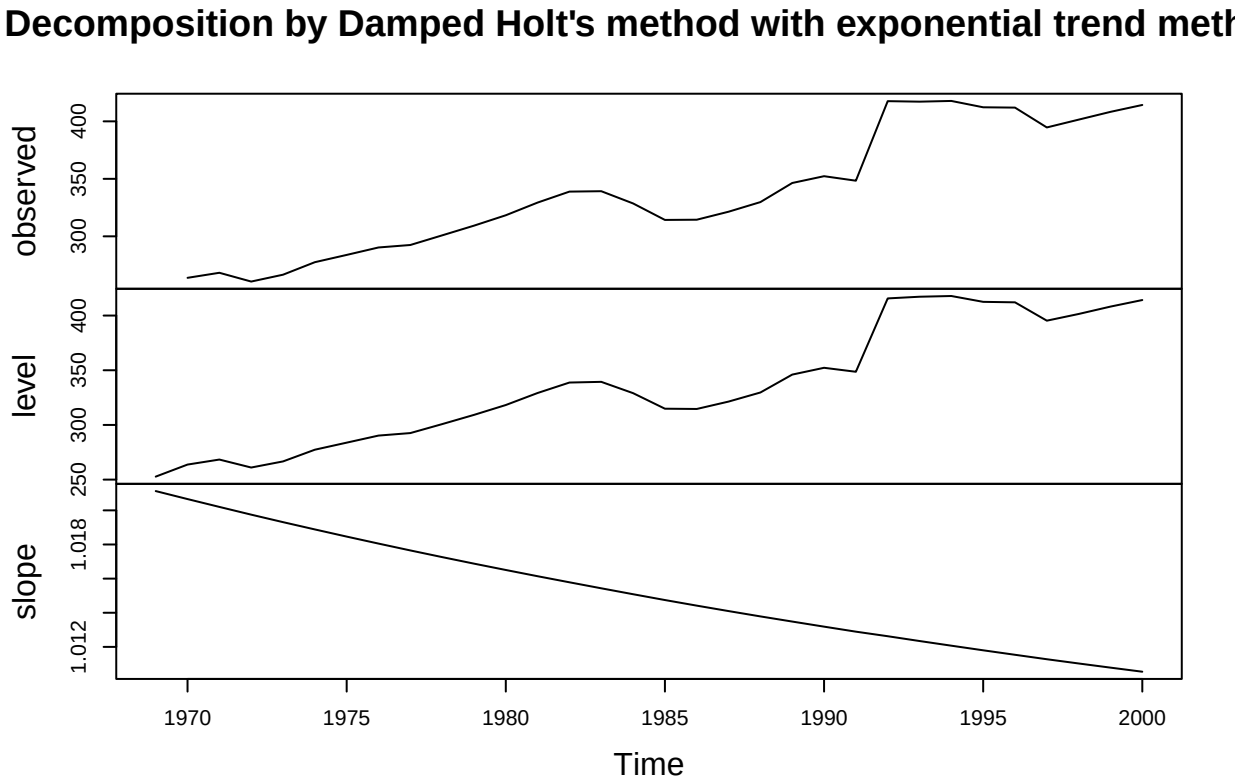


Interpretation

Comparing the Damped Holt forecast, the blue line, with the Test sample, the red line, we can see a slight difference with Simple Holt forecast. Now, at year 2004-2005, the difference between test sample and Damped Holt forecast seems bigger than with Simple Holt horecast, which suggest the forecast for that period has been damped too much.

5. Estimation by holt (exponential=TRUE, damped=TRUE) {forecast}, accuracy {forecast} in the training and test sample, and level and slope plot

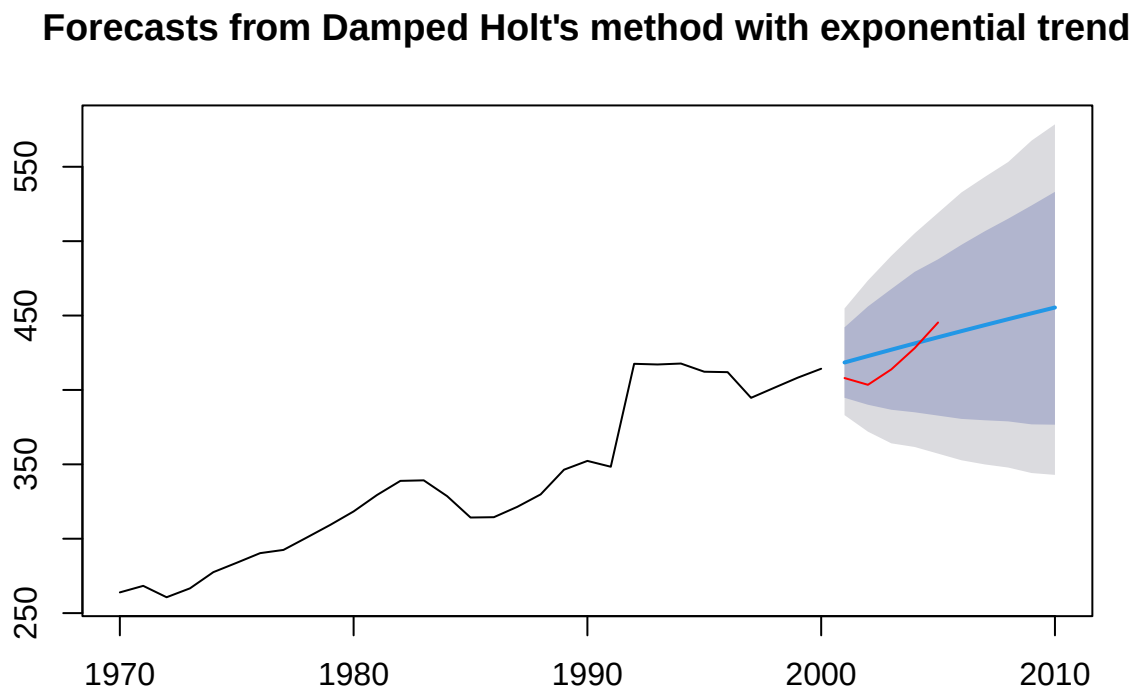
```
train_holt_damped_exp <- holt(train, damped = TRUE, exponential=TRUE)
plot(train_holt_damped_exp$model)
```




```r
accuracy(train_holt_damped_exp,test)
```

```
##                      ME     RMSE       MAE         MPE     MAPE      MASE
## Training set  0.3510945 14.04986  7.865819  0.03028087 2.233394 0.8725899
## Test set     -7.2441739 12.37620 11.215095 -1.81062344 2.702286 1.2441399
##                     ACF1 Theil's U
## Training set 0.002741395        NA
## Test set     0.311596444  1.042989
```

```r
plot(train_holt_damped_exp)
lines(test, col = "red")
```



Interpretation

Comparing the Damped Holt forecast with exponential trend, the blue line, with the Test sample, the red line, we can observe it is more alike to the Simple Holt forecast for this data.

6. Plot of the data and the forecasts of all the methods

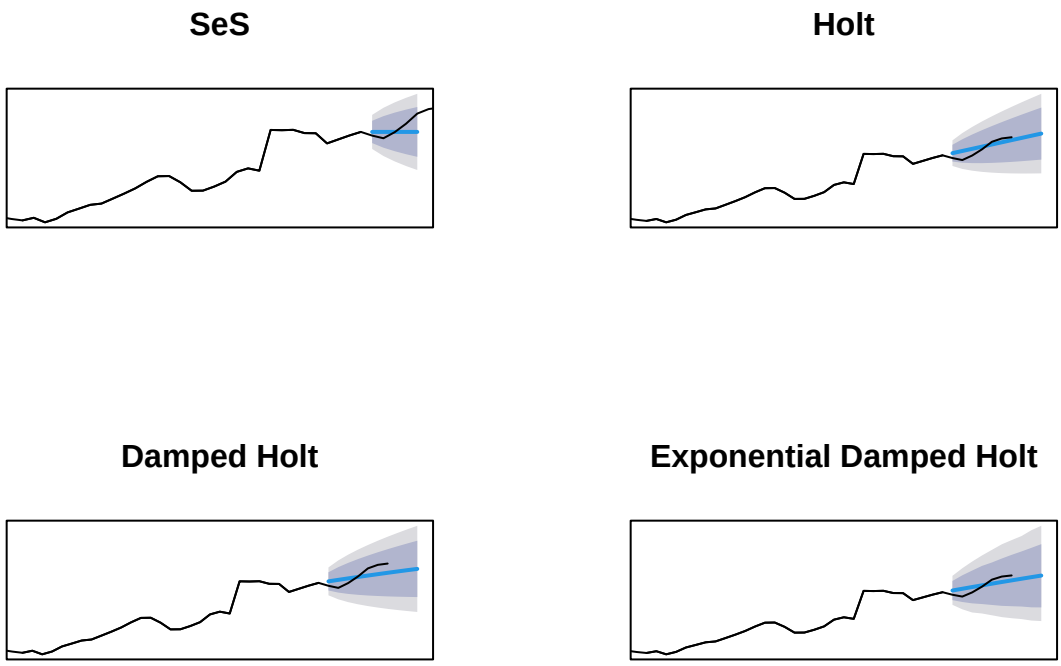
```
par(mfrow=c(2,2))

plot(train_ses, xaxt="n", xlab="", yaxt="n", ylab="", main="SeS")
lines(livestock,col="black")

plot(train_holt, xaxt="n", xlab="", yaxt="n", ylab="", main="Holt")
lines(livestock,col="black")

plot(train_holt_damped, xaxt="n", xlab="", yaxt="n", ylab="", main="Damped Holt")
lines(livestock,col="black")

plot(train_holt_damped_exp, xaxt="n", xlab="", yaxt="n", ylab="",
     main="Exponential Damped Holt")
lines(livestock,col="black")
```



Interpretation

Comparing all forecast methods, we could say Exponential Damped Holt or Simple Holt are the best fitted methods for the real data in this case, even though we should check for fitness in a larger horizon of time, as we do not have actual data from 2007 on. However, none of the methods are accurate for the first years of the forecast, as the actual data experimented a slightly decrease in 2001-2002 approximately.